A Template for IEEE Publications

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***Abstract*—This is the abstract. State the problem, the approach, and the main quantitative result in 150 to 250 words. IEEE abstracts are a single paragraph with no citations, footnotes, or undefined abbreviations.**

***Index Terms*—IEEE, IEEEtran, journal, conference, LaTeX, template**

I. Introduction

This template is a starting point for a manuscript submitted to an IEEE journal or conference; the venue type is set in `preamble/venue.tex`. Replace this text with the motivation for the work, a short review of prior art, and a clear statement of what is new. Most IEEE papers close the introduction with a numbered list of contributions and a one-paragraph outline of the rest of the paper.

The source is split so that co-authors can work on different parts at the same time. The title, author list, affiliations, and running header are defined once in `preamble/authors.tex` and shared by the main paper, the supplementary material, and the letters. Each section of the body is its own file under `sections/`, each table is its own file under `tables/`, figures go in `figures/`, and all references live in `bib/refs.bib`. The layout and the revision workflow follow the setup described in [1].

During a revision, wrap changed text in the macros from `preamble/revision.tex`: revised text shows up in blue, ~~deleted text is struck through in red~~, and changes made for a specific reviewer comment carry that comment's tag in the margin. **[TODO:** notes to co-authors look like this] All of them disappear in the clean copy.

R1.1

The rest of the paper is organized as follows. Section II reviews related work. Section III describes the proposed method, and Section IV reports the experiments. Section V discusses the results, and Section VI concludes the paper.

II. Related Work

Review the prior work closest to this paper, grouped by approach rather than listed one paper at a time. For each group, say what it does well and where it falls short, so the gap this paper fills is clear by the end of the section. The IEEEtran class itself is documented in [2].

A. First Line of Work

Summarize the first group of related methods and how the proposed method differs from them.

B. Second Line of Work

Summarize the second group of related methods and how the proposed method differs from them.

III. Proposed Method

Describe the proposed method here. Define every symbol at its first use and number every equation you refer to later, for example

$$y(t) = \int_0^t h(t-\tau)\, x(\tau)\, d\tau, \tag{1}$$

where $x(t)$ is the input, $h(t)$ is the impulse response, and $y(t)$ is the output. Refer to it as (1), or as Eq. (1) at the start of a sentence.

Cite prior work with `\cite`, as in the IEEEtran guide [2]. The `cite` package sorts and compresses multiple citations, so `\cite{a,b,c}` prints as a range. Add entries to `bib/refs.bib`; the IEEEtran bibliography style formats them in IEEE style.

A. Subsections

IEEE journals use up to three levels of headings: `\section`, `\subsection`, and `\subsubsection`. Keep each section in its own file so that `latexdiff` shows changes per section during a revision.

IV. Experiments

Give enough detail for someone else to reproduce the results, then report them. Put long parameter lists and extra results in the supplementary material and point to them from here (e.g., Section S-II).

A. Experimental Setup

Describe the datasets or test conditions, the baselines and how they were configured, the evaluation metrics, and the hardware and software used.

B. Results

Table I compares the proposed method against two baselines. Each table lives in its own file under `tables/` and is pulled in with `\input` where it is first mentioned. Table captions go above the table, in title case.

Figures work the same way, except that the caption goes below the figure. Save figures as vector PDF where possible

TABLE I
COMPARISON WITH BASELINE METHODS

Method	Accuracy (%)	Runtime (ms)	Params (M)
Baseline A	85.2	12.4	3.1
Baseline B	87.9	18.7	5.6
Proposed	**91.3**	**9.8**	**2.4**

and put them in `figures/`; `\graphicspath` is already set, so the file name alone is enough. Use `figure*` and `table*` for floats that span both columns.

V. DISCUSSION

Explain what the results in Section IV mean: why the method works, where it does not, and how sensitive it is to its main parameters. Compare against the related work in Section II where the numbers allow it.

A. Limitations

State the limitations of the method and of the evaluation honestly. Reviewers look for this, and it sets up the future work in Section VI.

VI. CONCLUSION

Summarize the contributions and the main results, and point out the limitations and the directions for future work. Do not introduce new results here.

ACKNOWLEDGMENT

The authors would like to thank XXX for helpful discussions.

REFERENCES

[1] X. Wang, "One prompt, one complete IEEE Transactions project with Claude Code," Blog post, Jul. 2026. [Online]. Available: https://blog.wangxm.com/2026/07/ieee-transactions-project-claude-code-one-prompt/

[2] M. Shell, *How to Use the IEEEtran LaTeX Class*, 2015, iEEEtran.cls version 1.8b. [Online]. Available: https://www.ctan.org/pkg/ieeetran